Equations of parallel and perpendicular lines

ANSWER KEY

Core Concepts: Equations of parallel and perpendicular lines

1 In $\triangle ABC$ studying equations of parallel and perpendicular lines, $m\angle A = 45^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 90° ✓

b. 100°

c. 82°

d. 135°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (45^\circ + 45^\circ) = 90^\circ$.

2 A geometric figure scaled by factor $k = 2$ has an original side length of 4 cm. What is the corresponding image length?

a. 6 cm

b. 8 cm ✓

c. 12 cm

d. 16 cm

Multiplying the original length by the scale factor gives $4 \times 2 = 8$ cm.

3 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 1$?

a. 4

b. 11

c. 9 ✓

d. 14

Substituting $x = 1$ into $y = 5x + 4$ gives $y = 5(1) + 4 = 9$.

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Procedural Fluency & Algebraic Manipulation

4 In $\triangle ABC$ studying equations of parallel and perpendicular lines, $m\angle A = 45^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 83° ✓

b. 93°

c. 75°

d. 135°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (45^\circ + 52^\circ) = 83^\circ$.

5 A geometric figure scaled by factor $k = 3$ has an original side length of 4 cm. What is the corresponding image length?

a. 9 cm

b. 12 cm ✓

c. 16 cm

d. 36 cm

Multiplying the original length by the scale factor gives $4 \times 3 = 12$ cm.

6 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 2$?

a. 9

b. 16

c. 14 ✓

d. 19

Substituting $x = 2$ into $y = 5x + 4$ gives $y = 5(2) + 4 = 14$.

Graphical & Geometric Interpretation

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7 In $\triangle ABC$ studying equations of parallel and perpendicular lines, $m\angle A = 45^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 76° ✓

b. 86°

c. 68°

d. 135°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (45^\circ + 59^\circ) = 76^\circ$.

8 A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length?

a. 12 cm

b. 16 cm ✓

c. 20 cm

d. 64 cm

Multiplying the original length by the scale factor gives $4 \times 4 = 16$ cm.

9 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 3$?

a. 14

b. 21

c. 19 ✓

d. 24

Substituting $x = 3$ into $y = 5x + 4$ gives $y = 5(3) + 4 = 19$.

Exam-Style Problem Solving & Modeling

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10 In $\triangle ABC$ studying equations of parallel and perpendicular lines, $m\angle A = 45^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 69° ✓

b. 79°

c. 61°

d. 135°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (45^\circ + 66^\circ) = 69^\circ$.

11 A geometric figure scaled by factor $k = 5$ has an original side length of 4 cm. What is the corresponding image length?

a. 15 cm

b. 20 cm ✓

c. 24 cm

d. 100 cm

Multiplying the original length by the scale factor gives $4 \times 5 = 20$ cm.

12 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 4$?

a. 19

b. 26

c. 24 ✓

d. 29

Substituting $x = 4$ into $y = 5x + 4$ gives $y = 5(4) + 4 = 24$.

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